Rooted-Tree Automorphism Groups
Wreath-product proof, exact matrix tests, and the limit boundary

Computational verification note

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Abstract

We prove the exact recursive sigma–MGF law for automorphisms of any finite rooted tree. Explicit leaf-permutation matrix groups for complete binary trees through height three and ternary trees through height two give 30 exact mixed-coefficient comparisons. The document separates this proved finite recursion from the still-open general random-tree Λ-limit.

1 Wreath-product lemma

Let a finite permutation group H act on X, and let $H \wr S_m$ act on $X \times [m]$. Define

$$\mathcal{M}_H(\mathbf{t}) = \frac{1}{|H|} \sum_{h \in H} \prod_i \det(I - t_i h)^{-1}.$$

Write an element of the wreath product as $(h_1, \dots, h_m; \sigma)$. On a top cycle $c = (i_1 \cdots i_\ell)$, one circuit accumulates $h_c = h_{i_\ell} \cdots h_{i_1}$ and block elimination gives

$$\det(I - tg|_c) = \det(I - t^\ell h_c).$$

Uniform independent h_i make the products h_c uniform and independent over the top cycles. Averaging over S_m and using its cycle index proves

$$\boxed{\sum_{m \geq 0} u^m \mathcal{M}_{H \wr S_m}(\mathbf{t}) = \exp\left(\sum_{\ell \geq 1} \frac{u^\ell}{\ell} \mathcal{M}_H(t_1^\ell, t_2^\ell, \dots)\right).} \tag{1}$$

2 Arbitrary rooted trees

If the root branches have isomorphism types A with multiplicities m_A, then internal automorphisms and permutations of equal branches give

$$\mathrm{Aut}(T) \cong \prod_A (\mathrm{Aut}(A) \wr S_{m_A}).$$

The leaf representation is the corresponding direct sum. Determinants multiply on direct sums, so (1) yields

$$\boxed{\mathcal{M}_T(\mathbf{t}) = \prod_A [u_A^{m_A}] \exp\left(\sum_{\ell \geq 1} \frac{u_A^\ell}{\ell} \mathcal{M}_A(\mathbf{t}^\ell)\right).} \tag{2}$$

For the complete b-ary tree, $G_h = G_{h-1} \wr S_b$, starting with the trivial action on one leaf.

3 Exact computational verification

The verifier recursively enumerates every wreath-product permutation and materializes its zero-one matrix. Distinct matrix keys and the identity

$$|G_h| = b!\,|G_{h-1}|^b$$

independently check each closure. For a matrix M, the direct symmetric characters are obtained over exact rationals from Newton's identity

$$dh_d(M) = \sum_{r=1}^{d} \mathrm{Tr}(M^r) h_{d-r}(M).$$

The other side is computed from the exact cycle-index average in (1), with truncated multivariate polynomial arithmetic. Thus the two routes share neither matrices nor coefficient extraction.

b	height	leaves	base order	group order
2	1	2	1	2
2	2	4	2	8
2	3	8	8	128
3	1	3	1	6
3	2	9	6	1296

For every row the program compares coefficients indexed by $(1), (2), (3), (1,1), (2,1), (2,2)$; all 30 comparisons pass exactly.

4 What is and is not proved

Equation (2) is an exact theorem for each finite tree. For a Galton–Watson tree it becomes a distributional recursion after conditioning on branch-type counts. Also,

$$\log|\mathrm{Aut}(T)| = \sum_i \log|\mathrm{Aut}(T_i)| + \sum_A \log(m_A!)$$

is an additive tree functional; Gaussian-type asymptotics for this scalar quantity are known. Neither fact alone proves convergence of the full random series $\mathcal{M}_T(\mathbf{t})$. A general limiting rooted-tree Λ-distribution therefore remains open, and the verifier makes no such claim.

5 Reproduction and reference

```
.venv/bin/python for_this_guy/limiting_lambda_claims/
  rooted_tree_automorphisms/verification.py
.venv/bin/marimo edit for_this_guy/limiting_lambda_claims/
  rooted_tree_automorphisms/notebook.py
```

Join wrapped commands onto one line. For automorphism-count asymptotics, see M. Olsson and S. Wagner, *The distribution of the number of automorphisms of random trees*, https://arxiv.org/abs/2211.12801.